

A Grammar of Graphics Approach to Managing Numerical Ties in Parallel Coordinate Plots

Denise Bradford

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1 Introduction and Motivation

Parallel coordinate plots (PCPs) are a powerful technique for investigating patterns across multiple attributes (variables) simultaneously (Inselberg 2009). PCPs assign each dimension of an n -dimensional dataset to a vertical axis, with the axes arranged in parallel (Wegman 1990). Observations are drawn as polylines connecting a single point along each dimensional axis in sequence, providing a perspective on hard-to-visualize multidimensional data that can be used to identify clusters, outliers, and other facets of a dataset.

When multiple observations share the same value in a given dimension, their polylines perfectly overlap at that changepoint, creating “visual collisions” that mask information about the joint distribution between variables and obscure the density along a single axis. The treatment of ties is an aspect not generally addressed in the original parallel coordinate plots of Inselberg (1985) and Wegman (1990).

PCPs were canonically used to show continuous random variables, but their generalization to categorical data only exacerbated the problem of ties. The most straightforward treatment of categorical random variables, which is taken by Categorical PCPs (Pilhöfer and Unwin 2013), is to assign each level a numeric value; this results in ties at any non-unique value for each variable; an example is Figure 2(a). [However, alternate treatments developed separately out of categorical data visualization research, including Parallel Sets (Kosara, Bendix, and Hauser 2006) and Alluvial diagrams, and the conceptually similar Sankey diagrams¹ Variations on these plots, such as common angle plots (Hofmann and Vendettuoli 2013), address perceptual distortions that arise when line widths are evaluated along the vertical rather than the orthogonal direction, offering an alternative encoding that more faithfully communicates categorical association strengths. Hammock plots (Schonlau 2003), another type of PCP that accommodates categorical and continuous variables, shows multiple bivariate relationships using parallelograms drawn between axes. Parallel Sets and Hammock plots preserve the marginal and bivariate frequency information, but do not allow reconstruction of the full multivariate density, because individual observations are not shown; because of the ties, CPCPs also do not allow for reconstruction of the full multivariate density, as it is impossible to distinguish a single observation as it passes through a cluster of observations on a categorical axis.

The development of generalized PCPs (via the software implementation `ggpcp` (VanderPlas et al. 2023)) represents a major step forward in representing the full joint distribution of the data while showing individual observations. One innovation in the `ggpcp` package is its handling of ties on *categorical* axes: each level is allocated a box proportional to its marginal frequency, and the observations sharing that level are spread evenly within the box so that individual lines remain visible. Sorting methods are implemented to distribute the tied values in a way that reduces the overall complexity of the plot, as random line-crossings can make PCPs difficult to read and interpret. Ties on an axis that remains *numeric*, however, are not spread: observations sharing a value are plotted at the same point and their lines overlap. This is the gap the present proposal addresses, and we return to it in detail in Section 4.

This project aims to extend the treatment of categorical variables in generalized PCPs by implementing a solution tie-breaking in continuous variables. The solution will have the following properties:

- marginal density information (approximately) faithfully represented on the vertical axis

¹Sankey diagrams do not always have parallel axes, but they are visually similar and frequently use parallel axis constructions.

- individual lines which can be visually distinguished, at least for data where the number of observations does not lead to overplotting
- observations will be sorted hierarchically within the tied treatment in order to minimize extraneous line-crossings that lead to perceptual complexity.

An immediate consequence of the second objective is that the viewer will be able to (at least in theory) reconstruct the full joint distribution between variables from the plot. We will extend the method used in `ggpcp` to represent categorical data to continuous variables, modifying it to maintain the approximate scaling of the continuous variable, with slight local distortions when tied values occur. In addition, we will develop a consistent visual representation to provide visual cues which indicate the presence of a local distortion due to tied values, while preserving the visual representation of singular values on the parallel axis.

This combination of the advantage of Hammock plots, maintaining the marginal density information, and generalized PCPs, which preserve individual observations and provide the ability to reconstruct the full joint distribution, will enhance GPCPs. Real-world data sets frequently have both categorical and numerical data, and duplicated values in either type of variable. Our extension of GPCPs will extend the ability to visually represent, examine, and assess these data sets.

1.1 Related Work

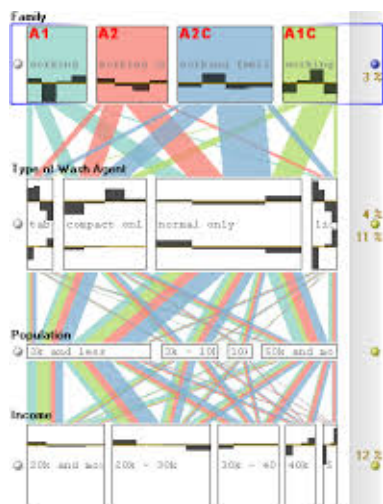
The parallel coordinate plot literature has developed over several independent avenues of investigation in different communities within statistics and computer science. Here, we review several historical attempts to address overplotting and the treatment of categorical variables.

1.1.1 Overplotting and density

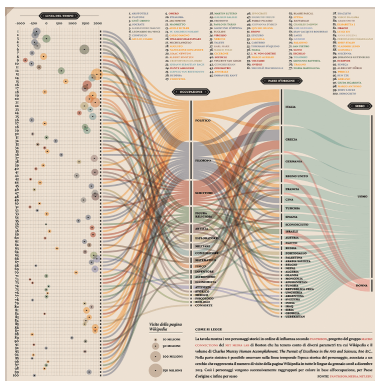
Early responses to visual overplotting in PCPs used transparency and kernel density estimation rather than positional adjustment. Alpha-blending, as used by Miller and Wegman (1990), allows the density of overlapping lines to become visible through saturation, but sacrifices the ability to trace individual observations. More recent approaches include density-based rendering (Heinrich and Weiskopf 2013) and edge bundling (McDonnell and Mueller 2008), both of which trade individual traceability for aggregate clarity.

1.1.2 Categorical extensions

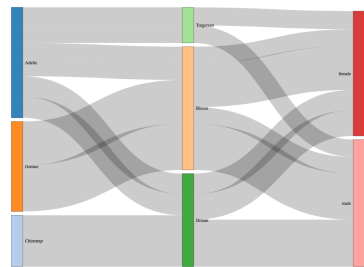
The categorical parallel axis plot literature developed largely independently of classical PCPs. Parallel Sets (Kosara, Bendix, and Hauser 2006) replaced individual polylines with ribbons whose widths encode bivariate frequencies, enabling unambiguous density reading at the cost of individual tracing (Figure 1a). Alluvial diagrams (Brunson 2020) and Sankey diagrams adopt a similar ribbon-based representation (Figure 1b and Figure 1c). Common angle plots (Hofmann and Vendettuoli 2013) addressed a further perceptual distortion specific to variable-width ribbons: the Müller-Lyer family of line-width illusions causes ribbon widths evaluated along the vertical to appear systematically different from their orthogonal widths. By constraining ribbons to a constant angle, common angle plots recover perceptually accurate frequency judgments (Figure 1d). Hammock plots (Schonlau 2003; Schonlau and Yang 2024) occupy a different position: they accommodate both categorical and continuous variables via constant-width parallelograms, but restrict inference to bivariate relationships between adjacent axes (Figure 1e).



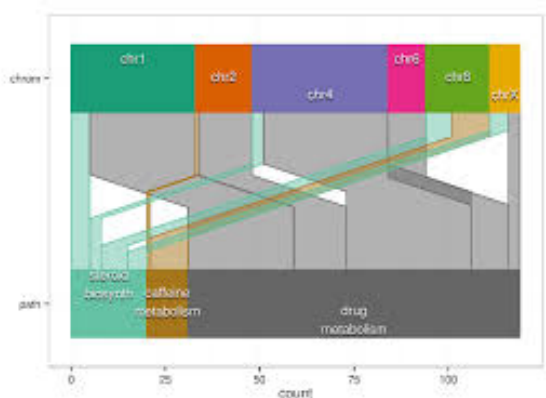
(a) A Parallel Sets display of several categorical variables.



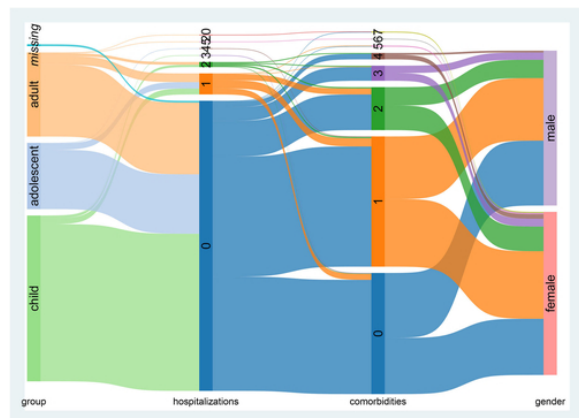
(b) An alluvial diagram, the Pantheon visualisation of the one hundred most influential historical figures tracing each figure across a timeline, occupation, country of origin, and sex.



(c) A Sankey diagram of the Palmer penguins data, with flows from species (Adelie, Gentoo, Chinstrap) to island (Torgersen, Biscoe, Dream) to sex.



(d) A common angle plot of the association between two categorical variables (chromosome and metabolic pathway).



(e) A hammock plot of the asthma data, with axes for age group, number of hospitalizations, number of comorbidities, and gender.

1.1.3 Perceptual bounds on vertical displacement

A separate and longstanding thread of the perceptual literature is potentially relevant to the question of how much spread the tie-breaking algorithm may safely introduce, though it is not intrinsic to the algorithm itself. The sine illusion, formalized by Day and Stecher (1991) and analyzed in the context of statistical graphics by VanderPlas and Hofmann (2015), establishes that the vertical extent of a line segment is not perceived neutrally: when segments of equal length are arrayed along a curve, those at extrema appear systematically longer than those at intermediate positions. The illusion belongs to the Müller-Lyer family that also underwrites the line-width illusion treated by Hofmann and Vendettuoli (2013) in their development of common angle plots. This is a concern that long predates the present work and arose independently of any tie-breaking method: common angle plots engaged with it as early as 2013 (Hofmann and Vendettuoli 2013), well before the generalized-PCP machinery this proposal builds on. We therefore do not treat it as a design constraint on the band-width parameter σ at this stage. Whether the sine illusion meaningfully affects within-band displacement here is an open empirical question, there are reasons to expect that continuous, traceable lines may largely avoid it, and we will need empirical tests before any perceptual bound on σ can be justified.

1.1.4 Axis and category ordering

A well-studied property of PCPs is that axis order strongly affects both the number of line crossings and the interpretability of patterns (VanderPlas et al. 2023). Minimizing line crossings through reordering is NP-hard in general (Martí and Laguna 2003), but heuristics based on correlation structure and sum of ranking differences provide tractable approximations (Ipkovich, Héberger, and Abonyi 2021). Within a fixed axis ordering, the order of factor levels and the plotting order of individual observations interact with tie-breaking: factor levels ordered to minimize crossings produce fewer extraneous crossings for any given tie-resolution strategy. Rendering order also matters: the `overplot` parameter in `ggpcp` controls which observations are rendered on top when lines overlap (VanderPlas et al. 2023). These considerations interact directly with the hierarchical sorting approach from `ggpcp` which has been extended in this proposal. This approach is discussed further in Section 4.3.

1.1.5 Computational quality metrics

Dennig et al. (2021) formalized a suite of screen-space quality metrics for Parallel Sets, including ribbon overlap, crossing angle, orthogonality, and ribbon width variance; the suite is collectively named ParSetgnostics. These metrics provide an empirical basis for comparing and optimizing parallel axis layouts without requiring user studies, and are candidate measures for the computational benchmarking component of our evaluation plan.

2 Categorical Variables and Ties

The `ggpcp` package currently addresses categorical ties through a combination of sorting and tie-breaking algorithms. The package implements hierarchical sorting through the `pcp_arrange(data, method, space)` function, which orders observations based on a hierarchical application of variable values. The `method` parameter determines the sequence in which variables are considered when resolving ties in the arrangement. The `space` parameter specifies the proportion of the y-axis dedicated to empty space between levels of categorical variables (Figure 2).

```
# 1. Modular Data Pipeline (The correct Step 1, 2, 3)
# Note: 'pcp_data' is used here as a variable name for the transformed object.
pcp_data <- mtcars %>%
  mutate(across(c(cyl, am, gear, carb), as.factor())) %>%
  pcp_select(cyl, am, gear, carb) %>%      # Step 1: Reshape data
  pcp_scale(method = "uniminmax") %>%      # Step 2: Scale axes
  pcp_arrange(method="from-left")          # Step 3: Break Ties,
                                          #         sorting from left
```

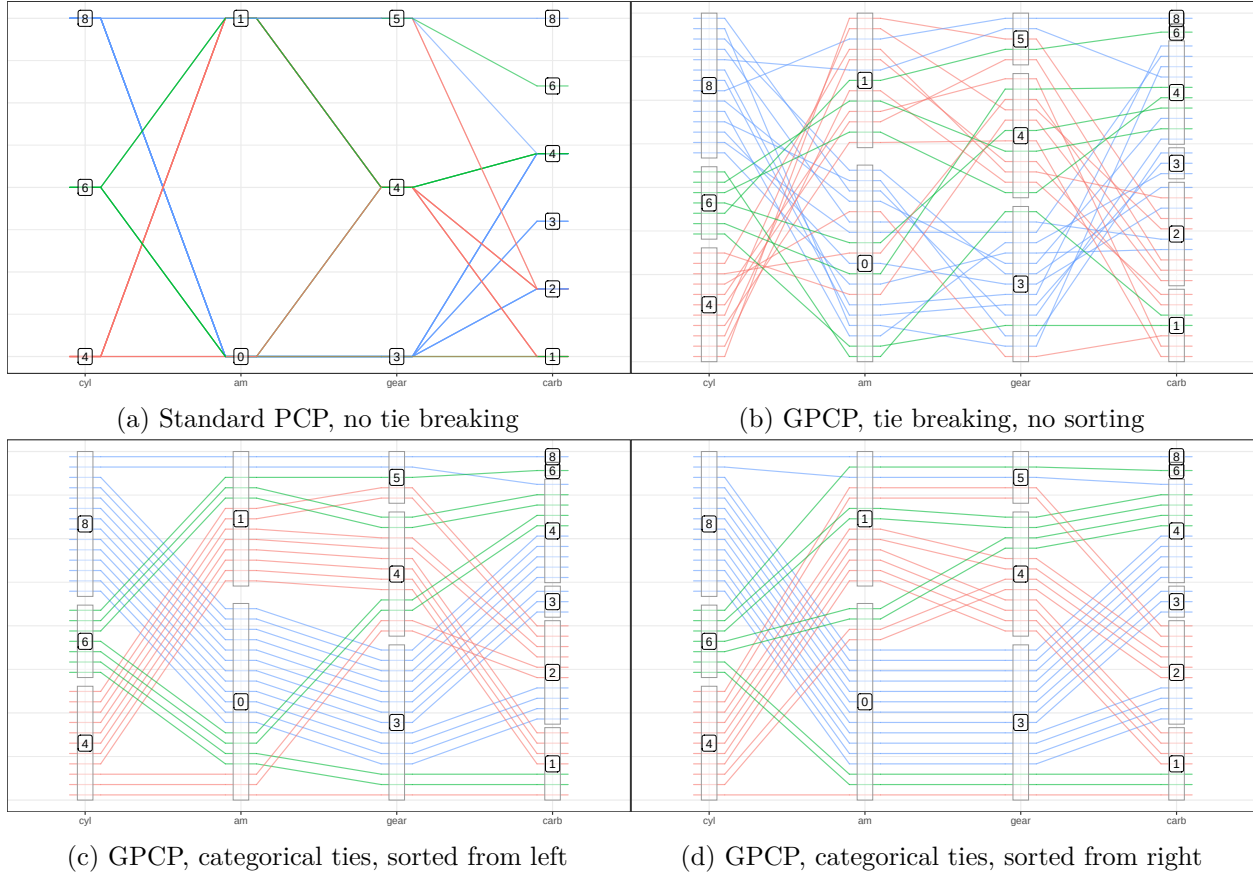


Figure 2: Comparison of tie-breaking methods for categorical variables in parallel coordinate plots using mtcars data. Standard PCP (top-left) shows overlapping lines without tie-breaking. GPCP with tie-breaking but no sorting (top-right) spreads observations evenly within categories. GPCP with sorting from left (bottom-left) and from right (bottom-right) apply hierarchical sorting to minimize line crossings, with light gray boxes indicating category groupings.

The ability to follow individual observations is key to being able to reconstruct the full joint distribution across parallel axes. `ggpcp` creates equally spaced points within each category that span the portion of the vertical axis dedicated to the category (preserving the marginal frequency information, which is optionally emphasized with `geom_pcp_boxes`). Figure 2(a) shows a PCP without this categorical tie-breaking approach while Figure 2(b) shows a PCP with tie-breaking but without sorting. `ggpcp` goes one step further by using hierarchical sorting (from the left, right, or both) to minimize unnecessary line crossings induced by the treatment of tied categorical variables.

The importance of hierarchical sorting to the overall visual effect of the tie-breaking treatment of categorical variables is emphasized through the contrast between Figure 2(b) and Figure 2(c) and Figure 2(d), which show sorting from the left and right respectively. We define a necessary line crossing as an intersection between lines from different observations which is induced by the order of x-axis variables and scaled values, rather than by the treatment of categorical or numerical tie-breaking algorithms. Hierarchical sorting minimizes the number of necessary line crossings, minimizing visual clutter. The hierarchical sorting serves as a form of “external cognition,” reducing the cognitive load required to “untangle” and mentally group crossed lines.

Having addressed categorical ties as implemented in `ggpcp`, it is worthwhile to examine how a viewer would follow a line across a generalized PCP with both numeric and categorical variables to illustrate the additional challenges introduced when numerical ties are present.



Figure 3: The parallel axes form the structural framework of the visualization.

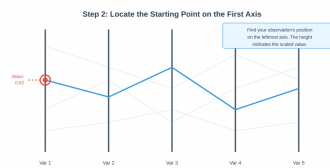


Figure 4: The starting point is identified on the first axis with its corresponding value.

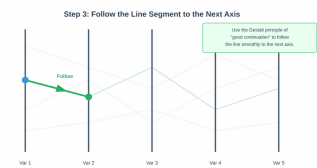


Figure 5: Following the line segment uses the Gestalt principle of good continuation.

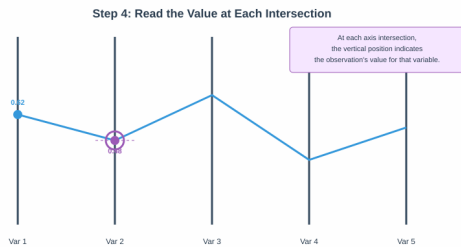


Figure 6: Values are read as approximate scaled value relative to the total range.

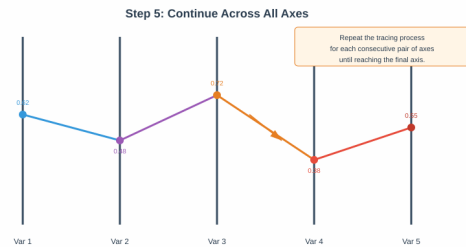


Figure 7: The complete traced path reveals the observation's values across all variables.

- Step 1: Identify the Parallel Axes (Figure 3).
Begin by identifying each vertical axis in the plot. Each axis represents one variable from the dataset. The axes are typically arranged from left to right, and the order may be determined by the data analyst to highlight specific relationships or minimize visual clutter.
- Step 2: Locate the Starting Point (Figure 4).
Find the observation of interest on the leftmost axis. The vertical position indicates the scaled value of that observation for the first variable. If you are examining a highlighted or color-coded observation, look for its distinctive marker at this starting position.
- Step 3: Follow the Line Segment (Figure 5).
Trace the line segment from the starting point to its intersection with the next axis. The human visual system naturally follows smooth, continuous paths due to the Gestalt principle of good continuation, which allows viewers to perceive connected line segments as a single, piecewise continuous line, making it easier to track observations across multiple parallel axes.

- Step 4: Read Values at Intersections (Figure 6).
At each axis intersection, the vertical position of the line indicates the observation's value for that variable. Read these values to understand how the observation changes (typically, relative to the other observations) across different dimensions of the data. The slope of line segments between axes provides information about the relationship between consecutive variables for that specific observation.
- Step 5: Continue Across All Axes (Figure 7).
Repeat the tracing process for each consecutive pair of axes until reaching the rightmost axis. By following the complete path, you obtain a comprehensive view of how that particular observation behaves across all measured variables. The vertical positions along each axis represent scaled values that can be interpreted as quantiles when the data are appropriately transformed. This enables identification of unique characteristics, cluster membership, or outlier status.

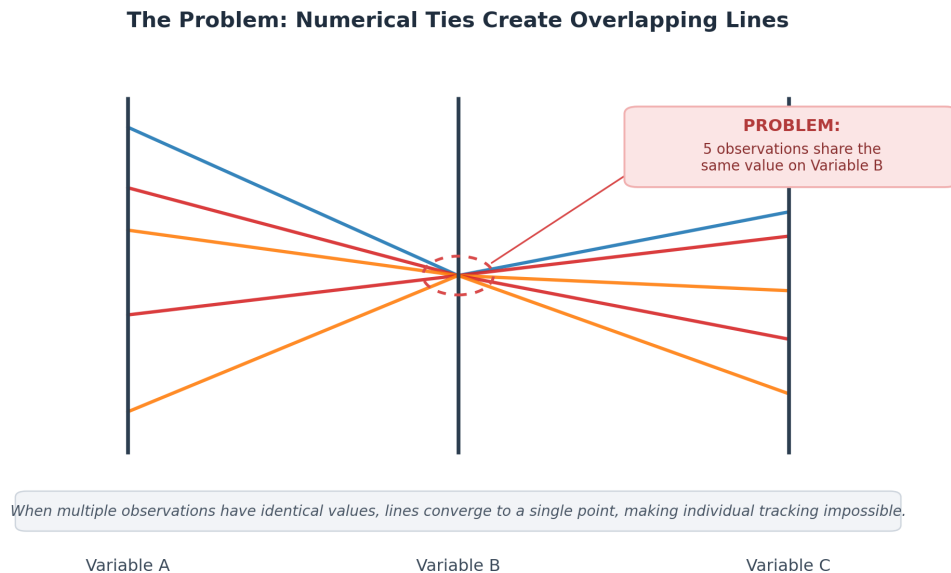


Figure 8: Numerical ties cause lines to converge at a single point, so individual tracking becomes impossible unless color is used to distinguish unique observations, as in this example. Even then, identity is recoverable only when lines arrive with sufficiently distinct colors: here the blue line is distinguishable by opacity, but the cyan and red lines are not cleanly separable, and the values each observation takes on Variables B and C are not uniquely determined from the plot alone.

- Step 6: Ties in Numbers — A Fork in the Road (Figure 8).

What was once a single point now becomes the starting point for two separate line segments that go in different directions. The Gestalt principle of good continuation doesn't work anymore when two or more lines come from the same point on an axis. In this configuration, color is the only channel that can provide separate line identity across the tie and even then, only when the palette is carefully chosen. The visual continuity that made tracing easy in earlier steps breaks down exactly where it is needed most. The observer cannot tell if the original observation follows the upper branch or the lower branch of the fork unless there are more visual cues, like color coding, clear labels, or a systematic tie-breaking arrangement. This lack of clarity is not just an aesthetic

problem; it also makes it harder for the user to identify patterns that would be difficult to see in tabular data by following individual observations through high-dimensional space. This ability to trace individual observations is one of the central reasons the improvements to PCPs in `ggpcp` are so effective.

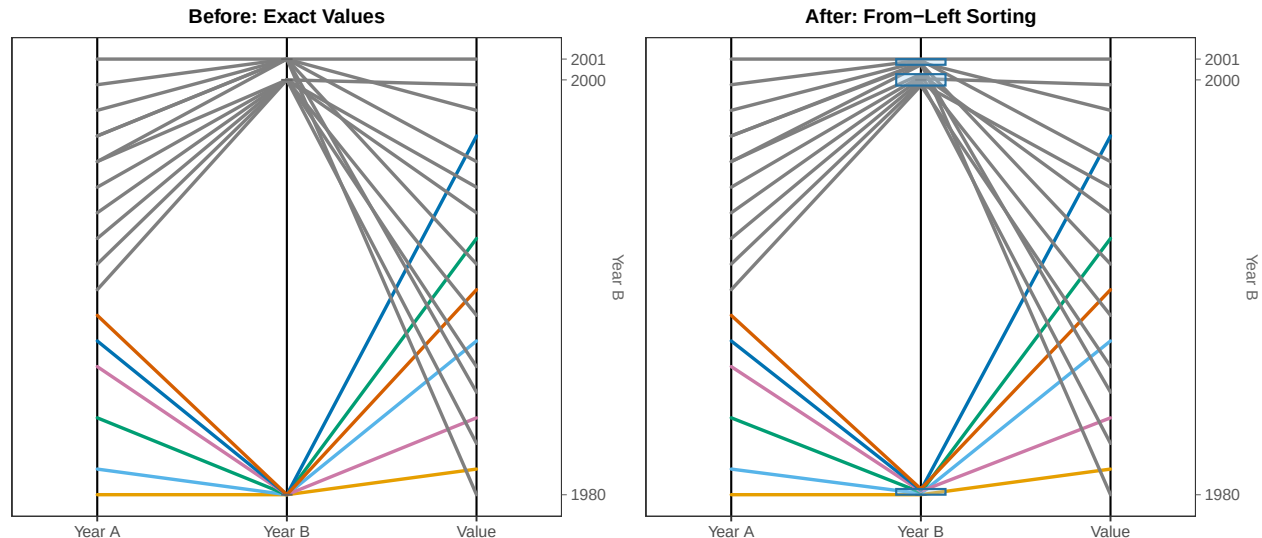


Figure 9: Comparison of a standard PCP with overlapping ties (left) and the proposed tie-breaking algorithm with **from-left** hierarchical sorting (right). Tied observations on Year B are spread within a resolution-based interval (blue boxes) and ordered by their Year A values, eliminating within-band crossings. A visible gap is preserved between the tie bands for 2000 and 2001: separation between bands is essential for the Gestalt grouping argument to hold, since adjacent bands that touch would visually merge and the reader would lose the ability to distinguish distinct values. The band width is therefore set strictly below the global axis resolution δ_j . Remaining crossings are necessary crossings induced by the data structure rather than artifacts of tie-breaking.

Two implementation details in Figure 9 deserve explicit comment because both are load-bearing for the Gestalt argument that motivates the development of a similar tie-breaking approach for numerical axes. First, each tie band must be narrower than the gap to its nearest neighbor: if the band around $v = 2000$ extended up to meet the band around $v = 2001$, the two values would visually fuse, defeating the separation the spread was meant to make visible. The band width in the figure is therefore set to a fraction of the variable resolution δ_j , leaving a deliberate gap between adjacent bands. Second, the reader must be able to *see* that gap, not merely have it exist in the coordinate system. This is a Gestalt-of-proximity requirement: the gap must lie above the just-noticeable-difference (JND) threshold and so cannot be shrunk to an infinitesimally small size. If ties at adjacent values are rendered without a perceptible gap, the grouping cue that “observations within a band share a value” competes with the unintended cue that “bands adjacent to one another belong to the same group.” The mathematical formalization of this separation requirement appears in Step 3 of the algorithm (Section 4.6.3.3). In practice, this restriction will be handled primarily through a warning printed by the software whenever the resolution becomes too small.

The `ggpcp` package implements a tie-breaking algorithm for categorical variables that maintains individual observation traceability by spacing observations evenly within each categorical level:

“All observations are spaced out evenly. This results in a natural visualization of the

marginal frequencies along each axis (additionally enhanced by the light gray boxes grouping observations in the same category) that is not as prominent in the previous three panels. The ordering of the observations within the level is such that a minimal number of line crossings occurs between the axes.” (p. 11)

This even spacing within an axis category allows the box height to function as a representation of the marginal proportion of the category relative to the whole. The sorting is hierarchical not as a stylistic choice but out of necessity: the levels must be processed in order for the arrangement to reduce line crossings across the plot as a whole. Without hierarchical sorting from the axis out to neighboring axes, lines departing toward different positions on the neighboring axis could cross *inside* the band.

This hierarchical sorting approach is the subject of Section 4.3; the spacing formula used for categorical levels is

$$d_i = \frac{S_i - S_i^- - S_i^+}{n_i - 1},$$

where:

- S_i is the total space allocated to category i
- S_i^- is the spacing below category i
- S_i^+ is the spacing above category i
- n_i is the number of observations in category i
- d_i is the optimal spacing distance between consecutive observations.

The spacing formula above is the categorical workhorse: each level i is allocated total height S_i , lower and upper margins S_i^- and S_i^+ are subtracted to enforce inter-level separation, and the remaining vertical extent is divided into $n_i - 1$ intervals to seat n_i observations. The proposed numerical framework adopts the same five quantities and the same formula structure, redefining only what determines each parameter; the full mapping is given in Section 4.6.4 (Table 2). Figure 11 displays the numerical version using the same parameter-key conventions as Figure 10, so the two figures can be laid side by side and read as a single procedure with type-specific instantiations rather than as two separate algorithms that happen to share an idea.

A key visual difference emerges when connecting categorical to numerical variables. As Schonlau and Yang (2024) observe in their comparison of the two displays, when many observations share a value on a categorical variable and on the adjacent numerical variable, the region spanning the two axes takes on a triangular shape, and the concentration of cases flowing from a single category into a small set of adjacent values reads far more readily in a hammock plot than in a GPCP.

Whether the transition takes a triangular or rectangular form is perhaps less important than what it communicates about the data itself. The more interesting question is whether the visualization preserves density information while still allowing viewers to follow individual observations. When observations flow from a categorical box toward numerical values, tie-breaking on the numerical axis shapes how clearly we can perceive the underlying distribution.

The `ggpcp` tie-breaking parameters, `method` and `space`, act on variables that are handled as categorical levels. To obtain any spreading on a *numeric* variable in the current package, a user must first coerce it to a factor. Doing so does spread the tied observations, but at a price: the metric scale is replaced by equally-spaced levels, so axis position no longer reflects magnitude or the gaps between

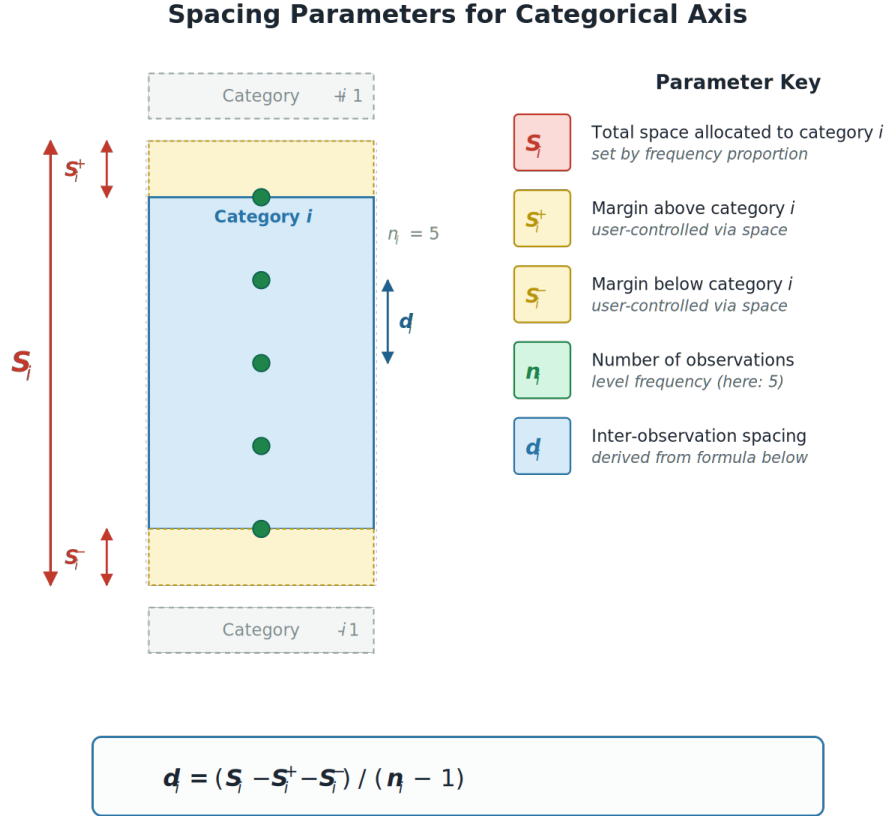


Figure 10: Spacing parameters for a categorical axis. Each level i is allocated total height S_i , with margins S_i^+ above and S_i^- below reserved for inter-level separation. The remaining vertical extent is divided into $n_i - 1$ intervals to seat n_i observations at uniform spacing d_i . The parameter-key panel on the right indicates how each quantity is determined: S_i is set by the level's frequency proportion under a user-chosen total budget, $S_i^\pm$ are user-controlled via the `space` argument, n_i is the level frequency, and d_i is derived from the formula at the bottom. The numerical analog of this figure (Figure 11) shares the same layout, color encoding, and formula structure; the differences are confined to how each parameter is set.

distinct values, and box height comes to encode frequency, so a heavily-tied value occupies more vertical space than a rarer value at a nearby magnitude. This represents a different set of trade-offs than hammock plots, which use constant-width parallelograms to show marginal densities directly while individual tracing requires highlighting. The proposed extension instead keeps the variable numeric: individual observation lines stay intact and density is conveyed by dot density within a resolution-anchored band, so the plot supports both seeing the forest and examining particular trees. We make this contrast precise in Section 4.1.

The preceding review of categorical tie-breaking, together with the related work in Section 1.1, sets the stage for a formal specification of the approach to numerical ties in this proposal detailed in Section 3.

3 Approach

This section states the proposal proper: the research questions the dissertation addresses, the contributions it claims, the plan for implementation and defaults, and the design of the evaluation. The technical development that these questions build on — the `tie_spread` algorithm, its mathematical specification, the visual-cue design, and preliminary visual results — is presented in Section 4. Section 4 should therefore be read as the developed method and prototype on which the proposal rests; the subsections below set out what remains to be built, defaulted, and empirically tested.

3.1 Research Questions and Contributions

The dissertation is organized around three research questions.

RQ1 (representation). Can tied observations on a numerical axis be spread so that, simultaneously, (a) the metric ordering and approximate spacing between distinct values is preserved, (b) the marginal density at each value remains recoverable, and (c) individual observations remain traceable across the plot? Section 4 develops a resolution-anchored construction intended to satisfy all three; the open work is to verify (b) and (c) empirically rather than only by construction.

RQ2 (defaults). What default value of the separation parameter σ , and what default visual-cue design, produce the best trade-off between the legibility of individual lines and the faithful perception of density and of the gaps between adjacent values? The provisional default $\sigma = 0.9$ is not yet empirically justified, and the perceptual bounds that would justify it (the just-noticeable-difference lower bound on the inter-band gap and any sine-illusion upper bound on within-band spread, Section 1.1) are open.

RQ3 (comparative value). Does numerical tie-breaking in GPCPs improve users’ frequency and comparison judgments relative to (i) GPCPs without it and (ii) hammock plots, the closest mixed-type competitor, and does it do so without sacrificing the individual-tracing advantage that motivates GPCPs?

The proposed contributions are correspondingly: (1) the `tie_spread` algorithm and its formal unification with the categorical tie-breaking already in `ggpcp` (Section 4.6.4); (2) a `tidyselect`-grammar implementation that lets users apply spreading per-axis while preserving exact positioning on others; (3) a visual-cue grammar, the numerical analog of `geom_pcp_boxes` and `geom_pcp_labels`, shared across categorical and numerical axes; (4) empirically grounded defaults for σ and the cue design; and (5) a controlled evaluation against hammock plots and un-broken GPCPs.

3.2 Implementation and Sensible Defaults

The grammar of graphics is built on the idea of smart defaults that can be customized when necessary. In `ggpcp`, categorical tie-breaking is supported by `geom_pcp_boxes` and `geom_pcp_labels`, which supply sensible visual cues for categorical levels. Two parameters require principled defaults for the numerical case: the separation parameter σ , which governs how much of each resolution band is used for spreading versus reserved as an inter-band gap; and the visual-cue rendering, including whether tie boxes are drawn, whether they are labeled with the original value, and whether boxes are suppressed below a tie-count threshold to reduce clutter.

The provisional default $\sigma = 0.9$ reserves 10% of the resolution band as a gap. It is provisional precisely because the two perceptual bounds that bracket a defensible value are not yet established: the gap must remain above the just-noticeable-difference threshold so that adjacent bands are not read as a single value (Section 4.6.3.3), while the spread should remain small enough that within-band displacement is not misread as genuine variance. We will fix σ with a calibration study (described with the main evaluation in Section 3.3) that varies σ over a grid at several axis resolutions δ_j and tie multiplicities k , and selects the largest σ — the most spread, and therefore the best individual legibility — that keeps band-separation errors below a pre-set threshold.

Implementation will proceed in three phases. Phase 1 builds the core `tie_spread` algorithm and the per-axis `tidyselect` interface, and resolves whether the tie-box geom is best implemented as a standalone layer or as a `ggproto` extension inheriting from the existing categorical box geom (Section 4.5). Phase 2 adds the numerical visual-cue geoms and the warning emitted when the resolution becomes too small for a perceptible gap. Phase 3 fits the defaults from the calibration study and documents them.

3.3 Evaluation: Comparison With Hammock Plots

Hammock plots (Schonlau 2003; Schonlau and Yang 2024) are the natural comparator: they are the closest mixed categorical/numerical parallel-axis method that foregrounds marginal density, and they make the opposite trade-off to GPCPs, encoding aggregate density through constant-width boxes while supporting individual tracing only through highlighting. Comparing GPCPs *with* numerical tie-breaking against GPCPs *without* it and against hammock plots isolates the contribution of the proposed method.

Tasks. We adopt four task types drawn from the elementary graphical-perception tasks used in the visualization literature: (1) *magnitude estimation*, how many observations take a given value on an axis; (2) *ordinal comparison*, which of two values, or which of two bands, holds more cases; (3) *ratio / proportion*, what fraction of the cases in a category flow to a given adjacent value; and (4) *tracing*, whether a highlighted observation takes the upper or lower value at the next axis, which probes the traceability claim directly.

Hypotheses. *H1*: on the density tasks (1–3), GPCP-with-tie-breaking is not worse than the hammock plot and both are better than GPCP-without, i.e. tie-breaking closes the density gap that motivates the work. *H2*: on the tracing task (4), GPCP-with-tie-breaking outperforms the hammock plot, since hammock tracing requires highlighting whereas the GPCP preserves continuous lines. *H3*: the advantage of tie-breaking on density tasks grows with tie multiplicity and is largest for integer-valued variables, where the converging-fan problem is most severe.

Design. The study is within-subjects on plot type (three levels: hammock, GPCP-without, GPCP-with) crossed with task type. Stimuli are generated from controlled synthetic datasets that vary

tie multiplicity, axis resolution δ_j , and the number of axes, supplemented by at least one real, integer-heavy dataset (for example the count variables in the penguins or Tour de France data) for ecological validity. Trial-level accuracy and response time are the outcomes, analyzed with mixed-effects models carrying random effects for participant and for stimulus. Presentation order is counterbalanced and attention checks are included. We will run a pilot to estimate response variance, fix the sample size to detect a medium effect at conventional power, and pre-register the design and analysis before data collection.

Computational benchmarking. Independently of the user study, we will compare layouts using the screen-space quality metrics of Dennig et al. (2021). Not all transfer: crossing count and crossing angle apply directly to individual-line GPCPs, whereas ribbon-overlap and ribbon-width-variance are defined for ribbon plots and have no direct analog. We therefore substitute line-level measures, overplot/line-overlap density and within-band crossing count, and use the latter to confirm computationally that hierarchical sorting removes within-band crossings as Section 4.3 claims, without requiring human subjects. This extends the qualitative box-shape comparison of Schonlau and Yang (2024) to a controlled perceptual experiment in which the tie-breaking extension is a distinct, manipulable condition.

4 Numerical Variables and Ties

Currently, numerical variables disrupt some of the best innovations in the GPCP approach by making it impossible to track an observation across the plot and by obscuring marginal density information through overplotting. The remaining two chapters of the dissertation will address the handling of ties on numerical axes and assess the utility of visual cues that can be paired with the tie breaking method to indicate that values are approximately accurate spatially and equivalent numerically.

The `ggpcp` package does not currently provide a mechanism for spreading tied values on a variable that remains numeric. We propose `tie_spread`, an algorithm that distributes observations with identical values along the vertical axis, transforming overlapping lines into a visually resolvable spread. Consistent with tidyverse conventions and the `tidyselect` grammar, the implementation allows users to specify tie-breaking behavior selectively across axes—applying spreading to some variables while preserving exact positioning on others. The following sections detail the design and implementation of this approach.

4.1 Relationship to Existing `ggpcp` Tie-Breaking

Because this proposal extends `ggpcp`, it is important to state precisely what `tie_spread` adds to behavior the package already has, since the two are easily conflated. The existing tie-breaking routine, `pcp_arrange`, is defined for variables that are handled as categorical levels: it allocates each level a box whose height is proportional to the level’s marginal frequency and spreads the observations evenly within that box, then orders them by hierarchical sorting (Section 2). A variable that is kept on a continuous scale receives no such treatment; it is plotted at its exact scaled value, so tied numeric observations coincide at a single point and their lines overlap (Figure 8).

There is an apparent workaround, and naming why it is inadequate is what makes the contribution precise. A numeric variable can be coerced to a factor, which triggers the categorical spreading above. This fails the numeric case on two counts. First, it discards the metric scale: distinct numeric values become equally-spaced factor levels, so vertical position no longer encodes magnitude and the

gaps between values — the very information a numeric axis exists to convey — are lost. Second, it makes box height encode frequency, so a value shared by many observations occupies more vertical space than a rarer value at a nearby magnitude, distorting the axis still further.

`tie_spread` changes exactly the part that fails. It spreads ties on a variable that stays numeric, anchoring every band to the global axis resolution δ_j so that band height is *constant* across values and the metric spacing of distinct values is preserved, and it encodes marginal density through *dot density within the band* rather than through band height. The hierarchical sorting that minimizes within-band crossings is inherited unchanged from the categorical machinery; what is new is the allocation and its semantics. The unified specification in Section 4.6.4 (Table 2) makes this exact: the spacing formula is identical in both cases, and only the right-hand column — how the per-value allocation $S(v)$ and the margins $S^\pm(v)$ are determined — differs from the categorical procedure.

4.2 The Problem: Overlapping Lines

When multiple observations share the same value on a numerical axis, their polylines converge to a single point, and the two failures noted in the introduction recur in their most acute form: individual traceability is lost, and marginal density is obscured (Figure 8). The problem is especially severe for integer-valued or coarsely-measured variables, where ties are structurally common rather than incidental.

The `tie_spread` algorithm resolves this by distributing tied observations within a vertical allocation anchored to the axis resolution δ_j — the smallest distance between any two distinct values on the axis — rather than within a frequency-proportional band. Equal margins are reserved above and below so that a gap is preserved between adjacent tied-value bands and the tied observations are spread evenly across the interior of the corresponding band. Figure 11 gives the parameters; the formal construction is set out in Section 4.6.3.3, and its correspondence to the categorical algorithm already in `ggpcp` is discussed in Section 4.6.4. The consequential design choice is what the allocation encodes: marginal density on a numerical axis is carried by the *dot density within each band* rather than by the *vertical extent of each band*. This preserves the natural numerical ordering on the axis and avoids the perceptual distortion that variable-height bands at fixed numerical positions would introduce.

Figure 12 shows the parameter spec of Figure 11 applied to a small parallel coordinate plot. Variable B carries three tied groups at $v \in \{0.8, 0.5, 0.2\}$, with $n(v) = 5, 3, 2$ observations respectively. All three bands have the same vertical allocation $S(v) = \delta_j$, regardless of n , the algorithm encodes marginal density through dot density within the band, not through band height. The figure illustrates **from-right** processing, the complement of the **from-left** mode used in Figure 9, so that observations within each band are placed in the order determined by their values on Variable C, the right neighbor. The polylines emerging from the band onto Variable C therefore carry no avoidable crossings; the within-band rank is the visible signature of the hierarchical sorting introduced in Section 4.3. Together the two figures illustrate that the algorithm is symmetric: the same construction applies under either processing direction, with the role of the reference axis swapped. The symmetry is worth stating because it is what makes a single implementation suffice for both **from-left** and **from-right**, and because the **from-both** mode is then simply the composition of the two — processing outward from a center axis, with each axis referencing its already-processed neighbor — rather than a third, separately-derived rule.

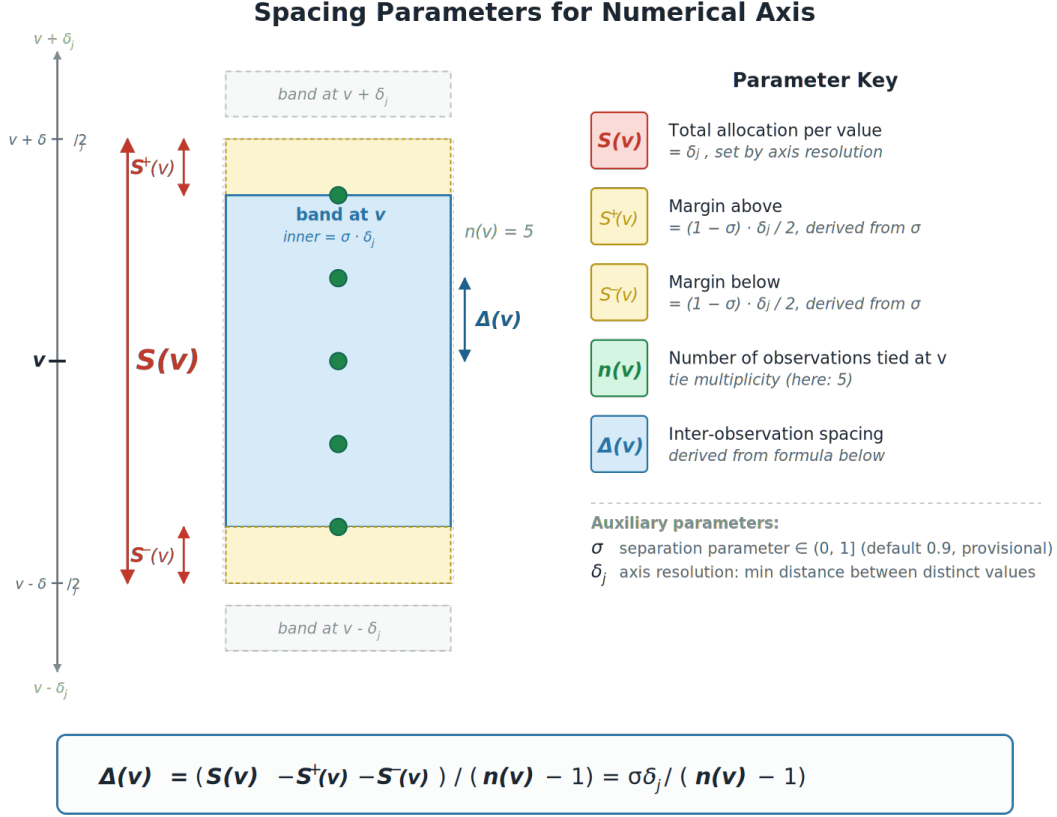


Figure 11: Spacing parameters for a numerical axis, displayed in the same layout as Figure 10 so the two frameworks read as a single procedure with type-specific instantiations. Each unique value v receives a vertical allocation of height $S(v) = \delta_j$, with margins $S^+(v)$ above and $S^-(v)$ below — both fixed at $(1 - \sigma) \cdot \delta_j / 2$ — that preserve the inter-band gap. The remaining inner region of height $\sigma \cdot \delta_j$ is divided into $n(v) - 1$ intervals to seat $n(v)$ tied observations at uniform spacing $\Delta(v)$. The mapping to Figure 10 is direct: $S_i \leftrightarrow S(v)$, $S_i^\pm \leftrightarrow S^\pm(v)$, $n_i \leftrightarrow n(v)$, $d_i \leftrightarrow \Delta(v)$, and the spacing formula at the bottom is structurally identical. The innovation, visible in the parameter-key annotations, is in how each quantity is determined: $S(v)$ is anchored to the global axis resolution rather than to frequency, and $S^\pm(v)$ are derived from the separation parameter σ rather than user-controlled by default.

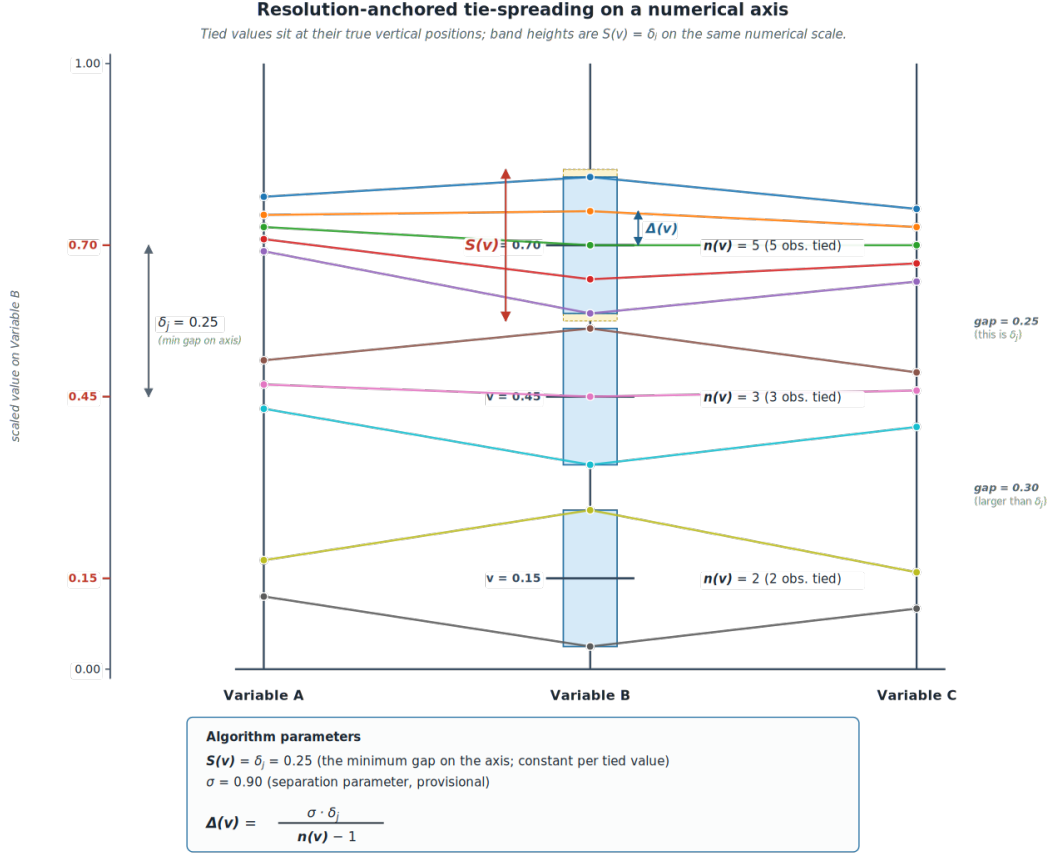


Figure 12: The algorithm of Figure 11 applied to a small parallel coordinate plot. Three tied groups on Variable B, with $n(v) = 5, 3, 2$, are spread within bands of identical vertical allocation $S(v) = \delta_j = 0.30$. The top band carries the parameter-reference annotations ($S(v)$, $\Delta(v)$, the yellow $S^\pm(v)$ margin strips); the mid and bot bands are shown as plain inner regions to keep the visual uncluttered. The solid horizontal line and “ $v = \cdot$ ” label at each band’s center mark the actual tied value, against which the within-band spread is the algorithm’s displacement of each tied observation. Within each band, the from-right ordering rule places the observation with the highest value on Variable C at the top of the band and the lowest at the bottom, so that polylines emerging from the band onto Variable C run in monotone order. Polylines are drawn as straight segments between axes; distinct colors per observation allow individual polylines to be traced through the tied region — the legibility property the algorithm is designed to preserve.

4.3 Hierarchical Sorting for Minimal Crossings

Spreading tied values evenly across a resolution-based interval is a necessary first step, but it is not, by itself, sufficient to eliminate visual knots. When the order of assignment within the tie band is arbitrary, lines departing toward different positions on the neighboring axis will cross inside the band, reproducing the very confusion the spreading was meant to resolve. The `ggpcp` package resolves this through hierarchical sorting that mirrors the approach already used for categorical variables.

For the `from-right` processing mode, the mode illustrated in Figure 13 and Figure 12, each tied observation $i \in T_j(v)$ is ranked by its adjusted position on the right neighbor axis, $\kappa_i = \tilde{y}_{i,A_j}$ (see Equation 9). Positions within the tie band are then assigned in the same rank order, from lowest to highest. This monotone rank-matching guarantees that no two lines cross within the tie band: the observation that arrives lowest on the right axis is placed lowest in the band, the next-lowest observation is placed next, and so on. Any crossings that remain after sorting are necessary crossings, those induced by the data structure itself rather than by the mechanics of tie-breaking.

By ordering observations within a tie group according to their values on adjacent axes, `ggpcp` produces locally parallel bundles of lines. Observations with similar trajectories are positioned in close proximity, so distinct groups are perceived as coherent bands moving together through high-dimensional space, an instance of the Gestalt principle of common fate. Common fate is usually defined in terms of motion, but even if the implicit ‘motion’ of following a line from axis to axis is not sufficient, the related principle of similarity of slope and line start and end points would suffice. In either case, the framing of common fate and similarity demonstrates why the ordering reduces visual noise and highlights underlying patterns in the data.

4.3.1 Interaction with axis ordering

The effectiveness of hierarchical sorting is closely intertwined with axis order and direction. VanderPlas et al. (2023) note that the order of factor levels is an important determinant of residual line crossings: some crossings are attributable to the data structure itself and cannot be eliminated by within-category sorting, while others arise from an unfavorable factor ordering and can be removed by reordering levels. Factor-level ordering is, helpfully, not an issue for numerical variables, there are no levels to reorder because numerical variables have an implicit quantitative order that is lacking in unordered categorical variables. Our tie-breaking extension therefore inherits a narrower dependency: for numerical axes, the minimum number of necessary crossings is a function of the scaled data values, the ordering of the axes, and the direction in which axes are processed hierarchically. Users should therefore consider axis ordering as a precondition for optimal tie-breaking, particularly when integer-valued variables with many ties are adjacent to continuous variables with high variance. `ggpcp` allows users to reorder factor levels via `mutate` before variable selection and to reverse axes to reduce negatively correlated crossings; the latter remains relevant for numerical axes.

4.4 Gestalt Principles and Line Continuity

The principle of good continuation states that the human visual system preferentially perceives smooth, continuous contours over interpretations requiring abrupt changes in direction. This principle directly impacts the effectiveness of parallel coordinate plots for tracing individual observations.

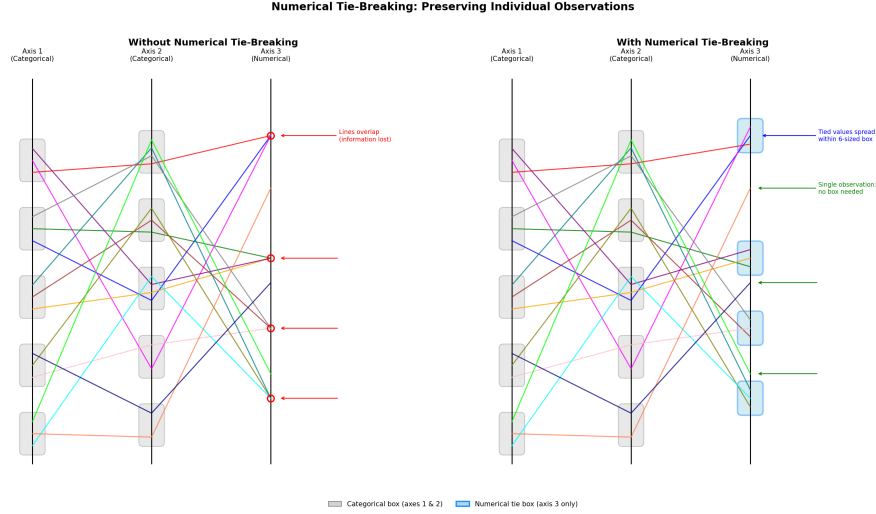


Figure 13: Comparison of parallel coordinate plots without (left) and with (right) numerical tie-breaking using **from-right** hierarchical sorting. When observations share the same numerical value, their lines overlap completely (left), hiding how many cases exist at that value. Spreading tied values within resolution-based boxes (blue) and sorting by the right neighbor axis (right) makes each observation visible, eliminates crossings within the tie band, and signals that positions within the box are artifacts of tie-breaking rather than meaningful data variation.

Equispaced lines maintain continuity by representing each observation as a continuous polyline extending from the leftmost to the rightmost axis. When observations share identical numerical values (ties), the equispacing mechanism distributes lines within each category so that they remain visually distinct without introducing discontinuities.

Constant-width boxes represent observations as aggregated area segments rather than individual lines. This encoding communicates aggregate quantities effectively: the area of each segment is proportional to the number of observations sharing that combination of values across adjacent axes, making the joint frequencies of two adjacent variables and axes immediately visible. By adding boxes to indicate tie-breaking repositioning along numerical axes, we emphasize the frequency information for the numerical variable. Without tie breaking, the user would have to infer the frequency information from the triangular “fan” shape of lines converging on a single point, but this requires additional cognitive load relative to a precomputed rectangle showing the marginal frequency directly.

In contrast, equispaced lines preserve individual observations as distinct visual elements. When lines are continuous and uninterrupted, viewers can leverage good continuation preattentively to trace a single case across all axes (Healey and Enns 2012). The two approaches thus serve distinct purposes: lines support individual-level tracing through low-level perceptual grouping, while boxes support aggregate-level frequency comparison.

4.5 Visual Cue Design

The introduction committed to developing “a consistent visual representation to provide visual cues which indicate the presence of a local distortion due to tied values.” This section specifies the design rationale for those cues, building on the perceptual principles outlined in the preceding section.

When tied values on a numerical axis are spread within a resolution-based interval, each line is displaced from its true value to a within-band position; that displacement is an artifact of tie-breaking rather than a reflection of meaningful data variation. A viewer who is unaware of this may misread the displacement as genuine within-group variance. The visual cue must therefore accomplish two things simultaneously: it must make the presence of a tie region legible without disrupting the tracing of individual lines through the region.

The design adopted here uses a light gray bounding box, anchored at the lower and upper bounds of the resolution interval $[L(v), U(v)]$, as illustrated in Figure 15 and Figure 16. This choice follows directly from the categorical tie-breaking convention already implemented in `ggpcp`, where gray boxes group observations within each categorical level. Extending the same visual element to numerical tie regions creates a unified grammar: a gray box in any axis region signals “these positions are spread for legibility, not because the underlying values differ.” This consistency reduces the cognitive overhead required to interpret mixed categorical-numerical plots.

Three alternative cue designs were considered and set aside. A color change on tied lines would require an additional channel already commonly used for group membership. A texture fill would introduce ambiguity on screen at small box heights. A dashed axis segment at the tied value would mark the location but not the extent of the spread, making it difficult to recover the original value. The gray box avoids all three problems: it marks both the location (the center of the box corresponds to the tied value) and the extent (the box height reflects δ_j , the global axis resolution), and it does not consume an encoding channel needed elsewhere.

The detection-and-rendering of these boxes can be expressed as a short pipeline that consumes the output of the tie-breaking step: identify values that appear more than once on a given axis, group contiguous spreads at numerical precision, and generate bounding-box coordinates for `geom_rect` rendering. An optional `label` argument displays the original tied value at the center of each box, which is particularly informative for integer-valued variables where many values may be tied at round numbers. (Whether this box geom is best implemented as a standalone layer or as a `ggproto` extension inheriting from the existing categorical box geom is taken up in Phase 1.)

4.6 Mathematical Framework

4.6.1 Notation

Let $\mathcal{D}$ be a dataset with n observations and p variables displayed on parallel axes $X_1, X_2, \dots, X_p$.

Table 1: Core notation

Symbol	Definition
x_{ij}	Raw value of observation i on variable j
y_{ij}	Scaled value (normalized to $[0, 1]$)
$\tilde{y}_{ij}$	Adjusted value after tie-breaking
$T_j(v)$	Set of observations tied at value v on axis j
k	Number of tied observations: $k = T_j(v) $
δ_j	Resolution of axis j : minimum distance between distinct values

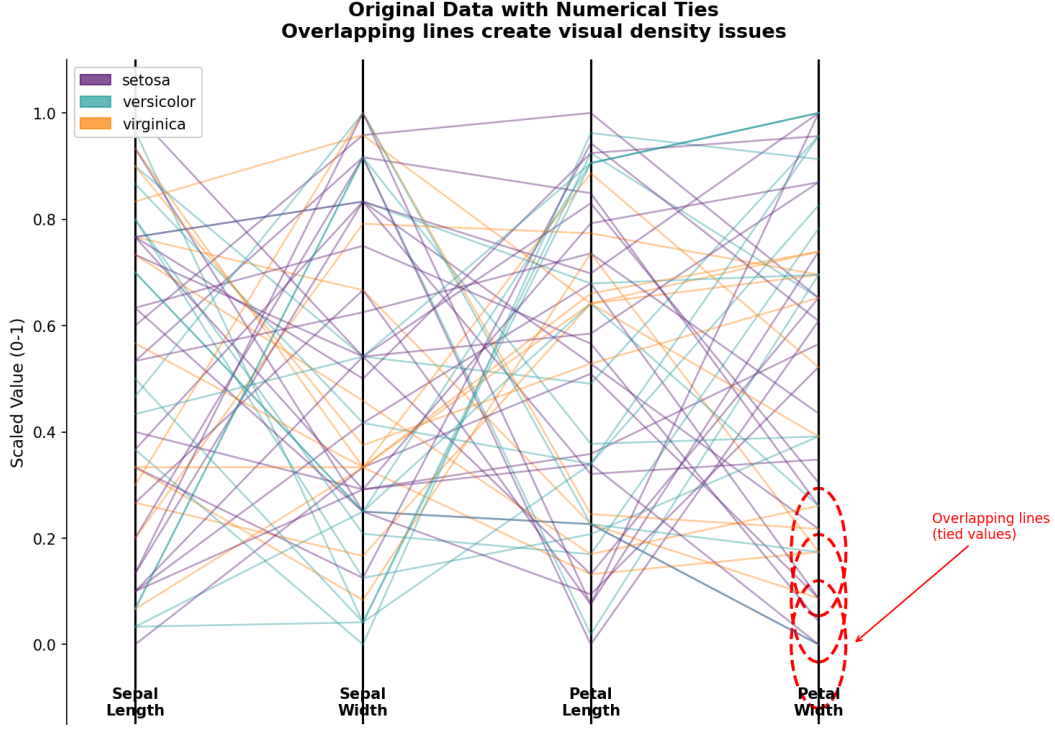


Figure 14: The tie problem in parallel coordinate plots. When observations share values, lines overlap and create visual collisions that mask the underlying data distribution.

4.6.2 Scaling

Variables are normalized to $[0, 1]$ using min-max scaling:

$$y_{ij} = \frac{x_{ij} - \min(X_j)}{\max(X_j) - \min(X_j)} \quad (1)$$

4.6.3 The Tie-Breaking Algorithm

4.6.3.1 Step 1: Compute Axis Resolution

To preserve density information and maintain consistent spacing, we first compute the global resolution of axis j —the smallest difference between any two distinct values:

$$\delta_j = \min\{|y_1 - y_2| : y_1 \neq y_2, y_1, y_2 \in Y_j\} \quad (2)$$

where Y_j is the set of unique values on axis j .

Using a consistent jitter range across all values ensures that the visual representation preserves both the density of observations at each value and the gaps between distinct values. An approach that computed jitter ranges individually for each tied value would risk collapsing visually meaningful gaps, destroying distance information in the display.

4.6.3.2 Step 2: Identify Ties

A value v on axis j is a tie if multiple observations share it:

$$T_j(v) = \{i : y_{ij} = v\}, \quad |T_j(v)| > 1 \quad (3)$$

4.6.3.3 Step 3: Compute Available Space

Each value v is allocated a vertical interval on the axis bounded by the half-resolution distance to its neighbors:

$$L(v) = v - \frac{\delta_j}{2}, \quad U(v) = v + \frac{\delta_j}{2} \quad (4)$$

If v lies within $\delta_j/2$ of an axis endpoint, the allocation is clamped to the unit interval so that no band extends past the plotted axis:

- $L(v) = \max(0, v - \delta_j/2)$
- $U(v) = \min(1, v + \delta_j/2)$

Away from the boundary, the total allocation per value is the full resolution:

$$S(v) = U(v) - L(v) = \delta_j \quad (5)$$

This $S(v)$ is the numerical analog of the categorical S_i : it is the full per-unit allocation on the axis, including the margins reserved for inter-unit separation. The clamped boundary case is the one place where the realized allocation $U(v) - L(v)$ is smaller than δ_j ; there the margins (Equation 6) and spacing (Equation 8) are computed from the realized allocation rather than from δ_j .

Separation constraint. Allocating the entire interval $[L(v), U(v)]$ to spread observations would, for evenly-spaced distinct values, produce regions that share endpoints with their neighbors and render as a single continuous column on the axis. This collapses the visual separation between distinct values and defeats the Gestalt grouping cue (see Figure 9). We therefore introduce a separation parameter $\sigma \in (0, 1]$ (default $\sigma = 0.9$, provisional) that reserves margins of equal width above and below the dot-placement region:

$$S^+(v) = S^-(v) = \frac{(1 - \sigma) \cdot \delta_j}{2} \quad (6)$$

These margins are the numerical analog of the categorical S_i^+ and S_i^- : where the categorical margins are user-controlled via the `space` argument, the numerical margins are derived from σ . The dot-placement region — the actual vertical extent within which tied observations are spread — is the residual:

$$S(v) - S^+(v) - S^-(v) = \sigma \cdot \delta_j \quad (7)$$

with bounds $[L(v) + S^-(v), U(v) - S^+(v)]$. The unencoded margins on either side preserve the perceptual gap between adjacent value bands.

4.6.3.4 Step 4: Compute Optimal Spacing

The inter-observation spacing is the dot-placement region (Equation Equation 7) divided into $k - 1$ intervals:

$$\Delta(v) = \frac{S(v) - S^+(v) - S^-(v)}{k - 1} = \frac{\sigma \cdot \delta_j}{k - 1} \quad (8)$$

This is the same formula used for categorical levels:

$$d_i = \frac{S_i - S_i^+ - S_i^-}{n_i - 1}$$

The two formulas are structurally identical; only what determines S and $S^\pm$ differs, frequency and the user budget for categorical levels, the axis resolution δ_j and the separation parameter σ for numerical values. The full correspondence is tabulated in Section 4.6.4.

4.6.3.5 Step 5: Order Observations Hierarchically

To minimize line crossings, tied observations are sorted by their positions on the adjacent axis before spread positions are assigned. For axis j , let A_j be the adjacent axis, the left neighbor under **from-left** processing, or the right neighbor under **from-right** processing.

For each observation $i \in T_j(v)$, the sorting key is:

$$\kappa_i = \tilde{y}_{i, A_j} \quad (9)$$

Observations are ordered so that $\kappa_{i_1} \leq \kappa_{i_2} \leq \dots \leq \kappa_{i_k}$.

This ordering is the step that resolves the visual “knots” illustrated in Figure 9. Spreading alone distributes tied points across the interval $[L(v), U(v)]$, but without a consistent rank-matching to the neighbor axis, lines departing the band will cross inside it. Sorting by κ_i before assigning positions in Equation 10 creates a monotone mapping: the observation placed at rank m in the tie band is exactly the observation ranked m -th on the neighbor axis. Under **from-right** processing, this means the lowest position in the band connects to the lowest position on the right axis, eliminating within-band crossings entirely. Any crossings that remain after this step are necessary crossings induced by the data structure itself.

Key insight: This requires *sequential* axis processing, the adjusted positions from previously processed axes inform the ordering on the current axis.

4.6.3.6 Step 6: Assign Positions

Given ordered observations $(i_1, i_2, \dots, i_k)$, positions are assigned within the dot-placement region, the interior of the allocation $[L(v), U(v)]$ after the margins $S^-(v)$ and $S^+(v)$ are reserved:

$$\tilde{y}_{i_m, j} = L(v) + S^-(v) + (m - 1) \cdot \Delta(v), \quad m = 1, 2, \dots, k \quad (10)$$

4.6.4 Unified Framework

The algorithm treats numerical values and categorical levels identically at the formula level: the same spacing equation governs both, and the same five quantities (S , S^+ , S^- , n , and the inter-observation spacing) are present in both cases. What differs is how each quantity is determined.

Table 2: Unified spacing framework: identical formula structure, type-specific parameter semantics.

	Categorical	Numerical
Total allocation	S_i (set by frequency proportion under user budget)	$S(v) = \delta_j$ (set by axis resolution; constant across values)
Margin above	S_i^+ (user-controlled via space)	$S^+(v) = (1 - \sigma) \cdot \delta_j / 2$ (derived from σ)
Margin below	S_i^- (user-controlled via space)	$S^-(v) = (1 - \sigma) \cdot \delta_j / 2$ (derived from σ)
Count	n_i (level frequency)	$n(v) = k$ (tie multiplicity)
Spacing	$d_i = (S_i - S_i^+ - S_i^-) / (n_i - 1)$	$\Delta(v) = (S(v) - S^+(v) - S^-(v)) / (n(v) - 1)$

The shared formula is **spacing** = (**allocation** − **margins**) / (**n** − 1). The innovation in extending the categorical procedure to numerical variables lies entirely in the right-hand column: anchoring the per-value allocation to the axis resolution and deriving the margins from a single perceptual parameter σ , rather than from frequencies and user input.

4.7 Visual Results

After applying the tie-breaking algorithm, observations that previously overlapped become visually distinguishable (Figure 15). Tie regions can additionally be annotated with boxes showing the original value and the spread region (Figure 16), and the `get_tie_box()` rendering pipeline is illustrated in Figure 17.

4.8 Algorithm Summary

Algorithm: Unified Tie-Breaking

Input: Data D , variables $X_1 \dots X_p$, method $\in \{\text{from-left, from-right, from-both}\}$

Output: Adjusted positions $_{ij}$

1. Scale all variables to $[0, 1]$
2. For each axis j , compute resolution: $\delta_j = \min |y_1 - y_2|$ for distinct $y_1, y_2 \in Y_j$
3. Set processing order based on method:
 - from-left: process axes $1, 2, \dots, p$; reference left neighbor
 - from-right: process axes $p, p-1, \dots, 1$; reference right neighbor
 - from-both: process outward from center axis; reference previously processed neighbor
4. For each axis j in processing order:
 - a. Identify tied values: $V = \{v : |T_j(v)| > 1\}$

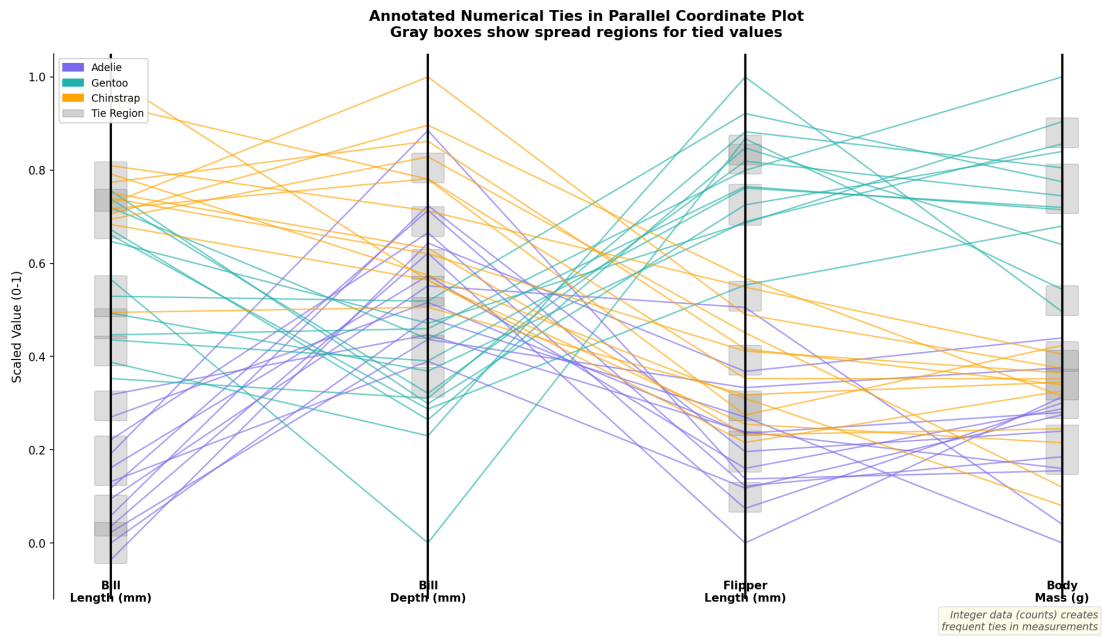


Figure 15: Parallel coordinate plot after applying uniform spacing. Gray boxes indicate regions where tied values have been spread, making individual observations traceable.

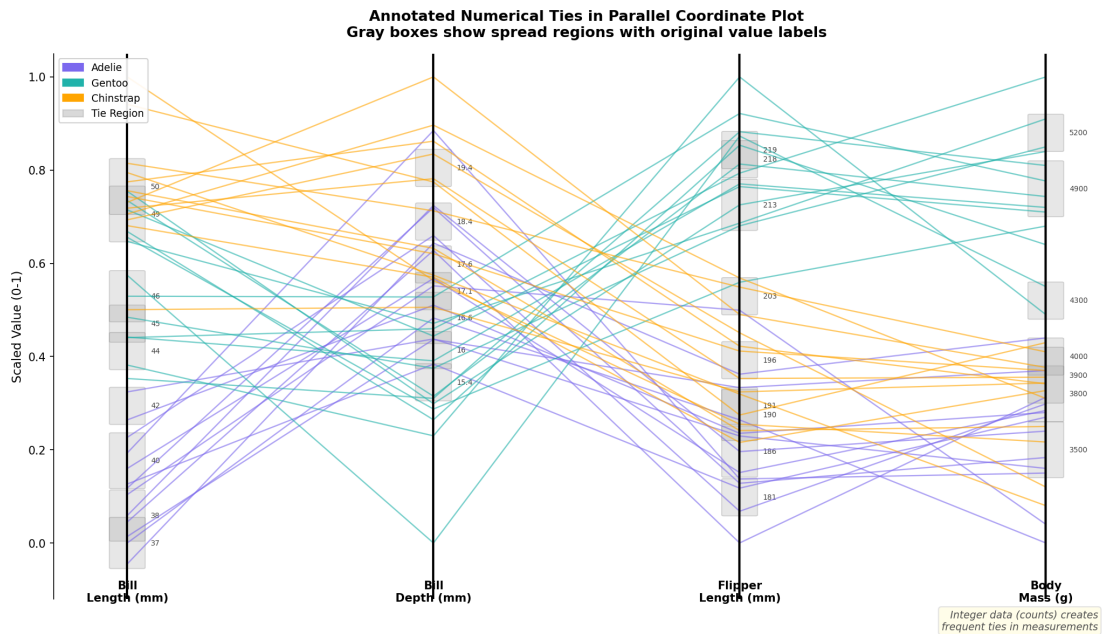


Figure 16: Annotated tie regions with labels. Gray boxes highlight where ties occurred, with labels showing the original tied value. This is particularly useful for integer data where ties are frequent.

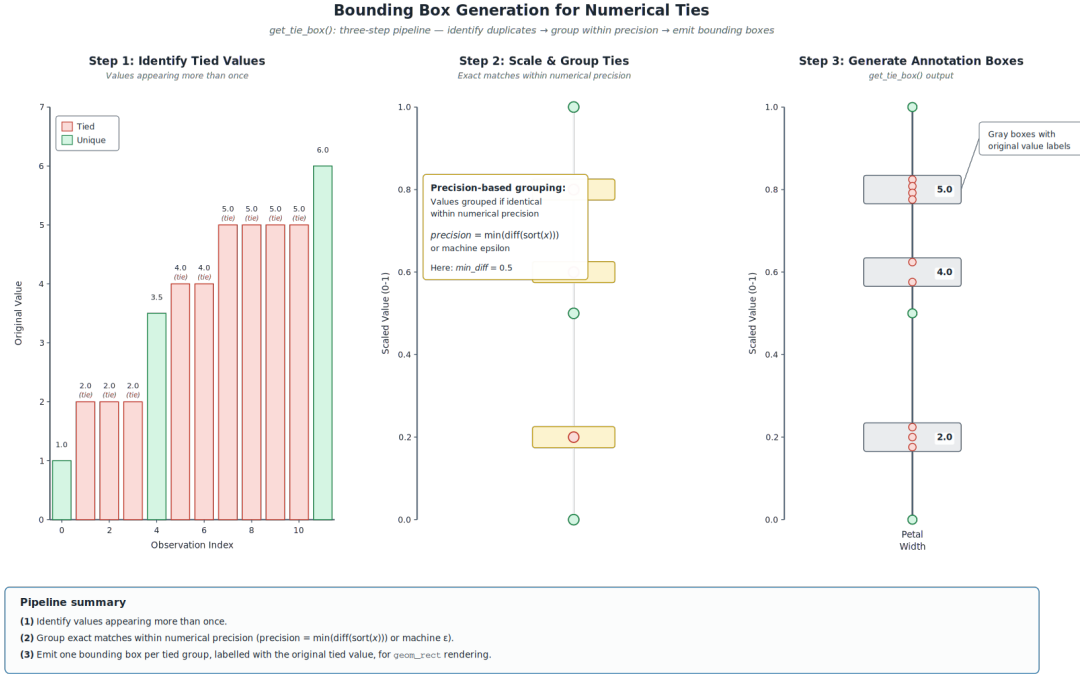


Figure 17: The tie box detection process. (1) Identify duplicated values, (2) scale and group contiguous spreads, (3) generate bounding box coordinates for annotation.

- b. For each $v \in V$:
 - i. Set total allocation: $S(v) = \delta_j$
 - ii. Compute outer bounds: $L(v) = v - \delta_j/2$, $U(v) = v + \delta_j/2$
 - iii. Set margins: $S^+(v) = S^-(v) = (1 - \sigma) \cdot \delta_j/2$
 - iv. Order $T_j(v)$ by $\kappa_i = \tilde{y}_{i,A_j}$ (positions on adjacent axis; right neighbor under **from-right**, left neighbor under **from-left**)
 - v. Compute spacing: $\Delta(v) = (S(v) - S^+(v) - S^-(v))/(k - 1)$
 - vi. Assign positions: $\tilde{y}_{i_m,j} = L(v) + S^-(v) + (m - 1) \cdot \Delta(v)$ for $m = 1, \dots, k$
5. Return adjusted data

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